Title: E-Commerce Funnel & Behavioral Analysis

Subtitle: Uncovering Customer Journey Insights from 99,000+ Real Orders

Your Name: Vimal Muthupandi

Tools Used: Python | Pandas | Matplotlib | Seaborn | Jupyter Notebook

What Business Questions Are We Solving?

Olist is a Brazilian e-commerce platform connecting small businesses to major marketplaces. With thousands of orders daily, understanding WHERE customers drop off and WHY deliveries fail is critical to growth.

Questions we set out to answer:

- How many orders successfully complete the full journey from placement to delivery?
- Where in the funnel are orders being lost?
- Which stage of the delivery process takes the longest?
- Which regions are experiencing the worst delivery performance?
- What is driving 1-star reviews and customer dissatisfaction?
- Which payment methods have the best conversion rates?

About the Dataset

Source: Olist Brazilian E-Commerce Public Dataset (Kaggle)

Time Period: 2016 – 2018

Total Orders: 99,441

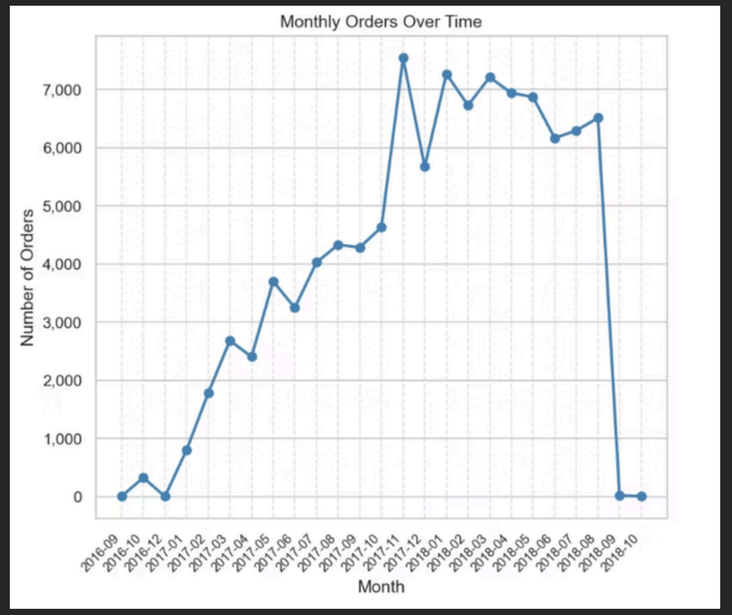
Total Revenue: R$16,008,872

Tables Used:

Table	Rows	What it Contains
orders	99,441	Order status & timestamps
customers	99,441	Customer city & state
order_items	112,650	Products, price, freight
payments	103,886	Payment type & value
reviews	99,224	Star ratings & comments
products	32,951	Category & dimensions
sellers	3,095	Seller location

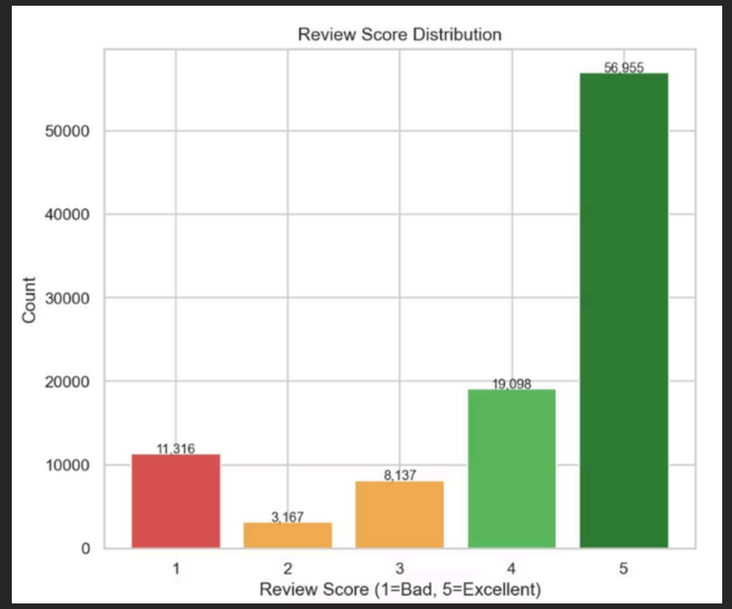
Approach: All 7 tables were cleaned, merged, and aggregated into one Master Table of 99,441 rows × 21 columns for analysis.

EDA Highlights



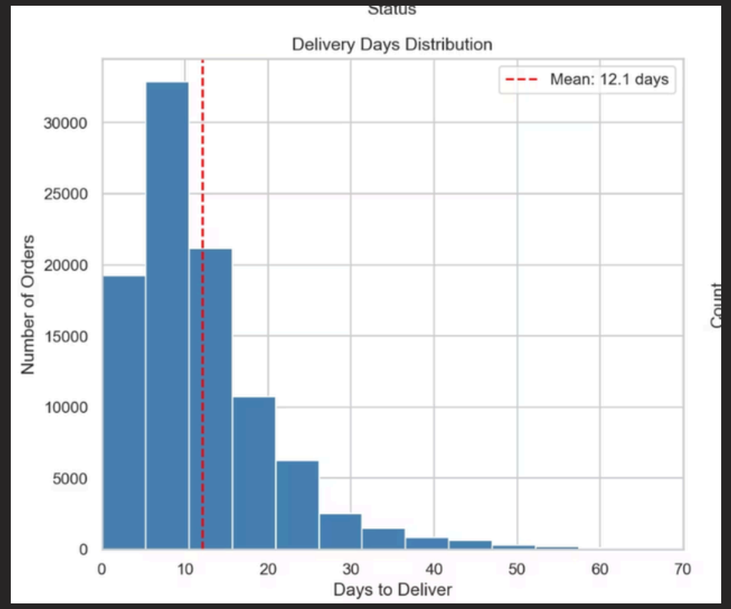
Monthly Orders Over Time

Orders grew 3× from mid-2017 to 2018, with a sharp Black Friday spike in November 2017 : confirming strong seasonal demand patterns.



Review Score Distribution

Most customers are satisfied (5-star majority), but a significant 1-star spike reveals a polarised experience. Unhappy customers are very vocal.

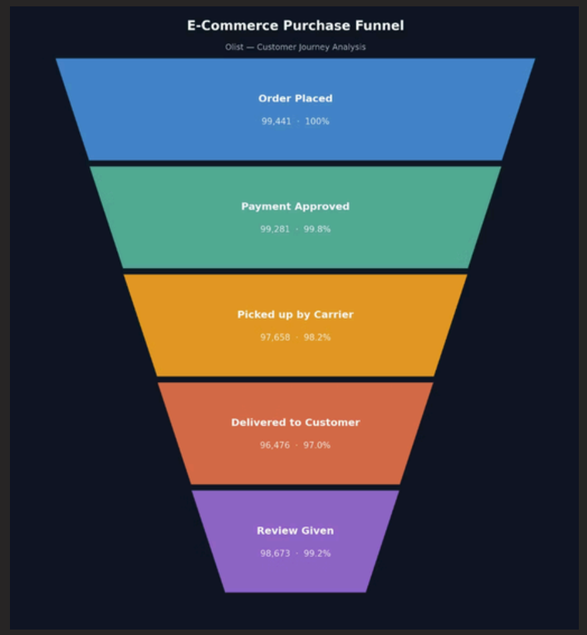


Delivery Days Distribution

Most orders arrive within 5–20 days (mean = 12.1 days), but a long right tail shows outlier orders taking 40 –60+ days : concentrated in remote states.

Funnel Analysis

Title: The Customer Journey — Order Funnel




```
Order Placed          →    99,441   (100%)
Payment Approved      →    99,281   (99.8%)
Picked up by Carrier  →    97,658   (98.2%)
Delivered to Customer →    96,476   (97.0%)
Review Given          →    98,673   (99.2%)
```

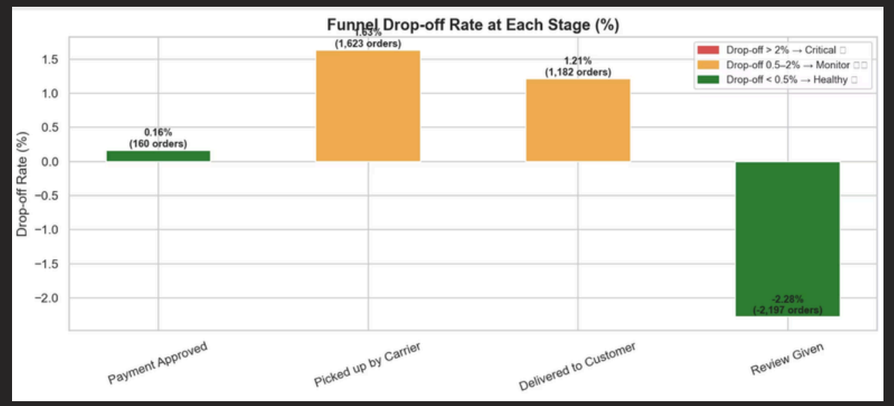
Key callout boxes to add:

Payment stage is nearly perfect : 99.8% approval rate shows a reliable payment system.

Carrier to Delivery sees the biggest absolute drop : 1,182 orders lost here due to logistics failures like lost packages or failed delivery attempts.

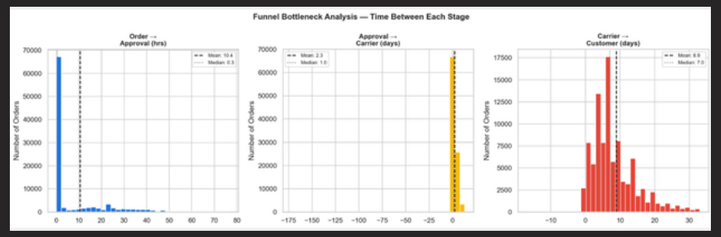
Overall conversion from Order Placed to Delivered = 97% : excellent for e-commerce.

Drop-off & Bottleneck Findings



Drop-off at Each Stage:

Transition	Orders Lost	Drop-off %	Status
Placed → Approved	160	0.16%	Healthy
Approved → Carrier	1,623	1.63%	Monitor
Carrier → Delivered	1,182	1.21%	Monitor
Delivered → Review	+2,197	—	Healthy



Time Bottleneck Between Stages:

Stage	Avg Time	Verdict
Order → Payment Approval	10.4 hours	Fast
Approval → Carrier Pickup	2.3 days	Seller prep time
Carrier → Customer Delivery	8.9 days	Biggest bottleneck

Key Insight to highlight: The carrier-to-customer leg alone accounts for 73% of total delivery time. **This is where the business needs to focus : better logistics partners or regional warehouses.**

Key Business Insights

Insight 1 : Strong Funnel Health 97% of all placed orders are successfully delivered. Olist's core operations are reliable. The payment system is particularly strong at 99.8% approval.

Insight 2 : Last-Mile Delivery is the #1 Bottleneck The carrier-to-customer stage takes 8.9 days on average , making up the majority of total delivery time. This is the single biggest area for improvement.

Insight 3 : Late Deliveries Are Killing Review Scores 7.9% of orders (7,827) arrived late. The 1-star review spike directly correlates with late deliveries. **Fix late deliveries → improve satisfaction scores.**

Insight 4 : Remote States Are Severely Underserved States like RR, AL, and MA have average delivery times exceeding 20 days and late rates above 15%. São Paulo receives orders in under 10 days on average. This gap is a major equity and growth problem.

Insight 5 : Credit Card Dominates & Converts Best 73% of transactions use credit card and it has the highest delivery conversion rate. Boleto (20%) requires manual payment , adding friction that can lead to order abandonment.

Recommendations

Recommendation 1 : Build Regional Warehouses in the North & Northeast States like RR, AL, MA, and PA have 20+ day delivery times. Placing fulfilment centres closer to these customers would cut delivery time by an estimated 40–50% and directly reduce late deliveries.

Recommendation 2 : Introduce Seller SLA Monitoring The approval-to-carrier stage takes 2.3 days on average. Enforcing stricter seller dispatch SLAs (e.g. ship within 24 hours of approval) would shrink total delivery time meaningfully.

Recommendation 3 : Send Boleto Payment Reminders Boleto payments require manual action by the customer. Automated SMS or email reminders before the boleto expires would reduce payment drop-off and recover lost orders.

Recommendation 4 : Flag High-Risk Orders Proactively Orders going to remote states, with low-rated sellers, or in heavy categories should be flagged early and rerouted to faster carriers , preventing late deliveries before they happen.

Recommendation 5 : Investigate & Fix the 1-Star Review Drivers Cross-reference all 1-star reviews with late delivery data, seller ID, and product category to pinpoint exactly which sellers and categories are causing dissatisfaction.

What's Next?

How This Analysis Can Go Further

Next Step 1 : Cohort Analysis Group customers by the month they first ordered and track their behaviour over time. Do early customers buy more frequently? Do customers who had a late first delivery ever return? Cohort analysis answers retention questions.

Next Step 2 : Seller Performance Scorecard Build a per-seller dashboard showing cancellation rate, average review score, average dispatch time, and late delivery rate. This would help Olist identify and remove underperforming sellers.

Next Step 3 : Interactive Dashboard Recreate this entire analysis in Tableau or Power BI so business stakeholders can filter the funnel by state, time period, category, or payment type without needing to run Python code

Thank You

This project demonstrates how raw transactional data can be transformed into clear business insights : identifying a 97% conversion funnel, a critical last-mile bottleneck, and actionable recommendations to improve delivery performance across Brazil.

Your details:

- Vimal Muthupandi
- 💼 https://www.linkedin.com/in/vimal-muthupandi/
- 🐙 https://github.com/VimalMuthupandi
- 📧 vimalm012003@gmail.com